DSAA3073 · Theories in Data Science

Choose a model, then evaluate the choice

Learning Sheet · Lecture 7

Cross-validation and model selection criteria

The choice is part of the learning procedure

Ridge regression gives us a fitted predictor once we specify a regularization parameter λ. The previous lesson explained how that parameter changes the fit. It left a practical decision: which value should we use when a new observation arrives?

The smallest training error answers a question about the observations already used in fitting. Our prediction goal concerns another observation. We therefore need evidence that tests a candidate away from the responses that determined its fit. There is a second issue. Once we use that evidence to choose a candidate, the evidence has also influenced the learning procedure.

LESSON BACKBONE

What evidence should choose a model, and what evidence should evaluate the completed choice?

We first design the data roles for fitting, selection and evaluation. We then calculate leave-one-out prediction losses, including an exact shortcut for fixed ridge regression. Finally, we compare likelihood criteria by their targets and conditions, and use the appropriate evidence to evaluate the complete choice.

A **candidate** here is a specified fitting rule with specified settings. For example, ridge with $\lambda = 2$ is one candidate and ridge with $\lambda = 10$ is another. Fitting a candidate uses data to estimate its coefficients. Selecting a candidate uses evidence to decide which fitting rule or settings to retain. These are different operations, even when the same computer program performs both.

We begin with squared prediction loss. If a predictor gives $\hat{f}(x)$ for a response y, its loss is $\big(y - \hat{f}(x)\big)^2$. The prediction risk of the fitted rule is the expected loss on a fresh observation from the population of interest. The word "fresh" matters: its response was not available when this rule was fitted.

Reserve observations for prediction

Definition: Holdout validation

Divide the available development data into a training set and a validation set. Fit each declared candidate using only the training set. Its **holdout validation score** is the mean loss on the validation observations. For squared loss,

$$\hat{R}_{\text{val}}\big(\hat{f}\big) = \frac{1}{n_{\text{val}}} \sum_{(x_i, y_i) \in S_{\text{val}}} \big(y_i - \hat{f}(x_i)\big)^2.$$

Here S_{val} is the validation set and n_{val} is its number of observations. The validation responses are used to score predictions, not to fit that candidate's coefficients.

For a fixed fitted candidate, independent validation observations from the same population have expected loss equal to its prediction risk. Taking their mean preserves that expectation. The observed mean still varies with the observations that happen to be available. It also concerns the candidate fitted on the training subset, not a different fit trained later on more data.

The following small example makes the two operations visible. It is a constructed comparison, not a claim about the accuracy of ridge regression in general.

Worked example: Training fit and held-out prediction can prefer different candidates

Use four training observations

$$(x_i, y_i) \in \{(1,1), (2,2), (3,2), (4,4)\}.$$

Fit the one-coefficient model $\hat{f}(x) = x\hat{\beta}_\lambda$, with no intercept and no estimated feature transformation. Its objective is

$$\sum_{i=1}^{4} (y_i - x_i\beta)^2 + \lambda\beta^2.$$

Two separate validation observations are $(5, 4)$ and $(6, 5)$. The calculated results are supplied below.

λ	$\hat{\beta}_\lambda$	Training mean squared loss	Validation mean squared loss
0	0.90000	0.17500	0.20500
2	0.84375	0.19873	0.02588
10	0.67500	0.55469	0.64656

Training loss prefers $\lambda = 0$. The validation comparison prefers $\lambda = 2$. The second choice allows a slightly worse training fit because it predicts these unused responses more closely. The table supplies a selection decision; it does not prove which candidate has the smallest population risk.

Keep the four training rows and the two validation rows distinct. We will return to the four training rows to examine a deletion calculation. The two validation observations will remain outside that calculation.

Rotate the role of the held-out observations

A single split may leave relatively few rows for fitting or relatively few for validation. It also makes the comparison depend on one particular assignment of roles. We can obtain predictions for more observations by changing which rows are held out. Each prediction must still come from a fit that excludes the row being scored.

Definition: k-fold cross-validation

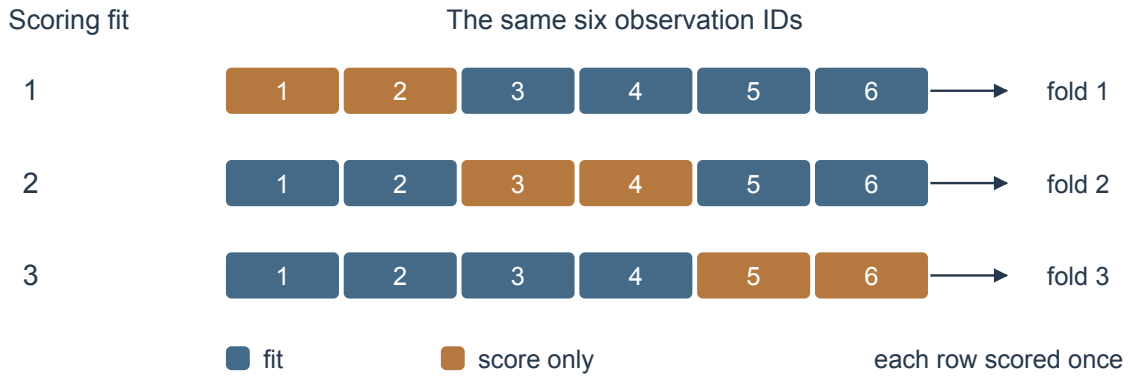
For n observations, partition their indices into k disjoint folds $I_1, ..., I_k$. For fold j, fit the candidate using all observations outside I_j; call this fit $\hat{f}^{-j}$. Score the observations in I_j with that fit. For squared loss, the overall score is

$$\mathrm{CV}_k = \frac{1}{n} \sum_{j=1}^{k} \sum_{i \in I_j} \left(y_i - \hat{f}^{-j}(x_i)\right)^2.$$

Every observation contributes one held-out prediction loss. The candidate settings stay fixed while the folds rotate.

There are two indices in this expression. The index j identifies the fold and hence the fitted model. The index i identifies an observation scored by that model. All observations in one fold

share the same fitting set. An observation's response can be used in other folds' fits, but never in the fit that produces its own scored prediction.



In each horizontal row, blue observations fit one model and amber observations are scored by it. Reading down a column shows that each observation is held out once. The numbers are observation IDs, not loss values.

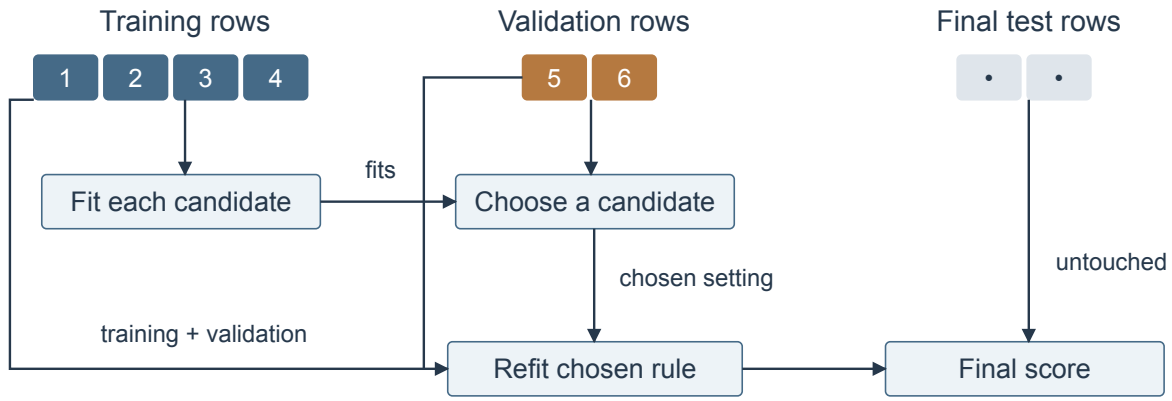
If all folds have equal size, averaging the k fold means gives the same score. If their sizes differ, fold j must receive weight $\frac{|I_j|}{n}$ to obtain the mean over observations. Giving every fold the same weight would give a row in a small fold more influence than a row in a large fold.

The choice of k changes both training-set size and computational work. It does not create a universal ordering of statistical variance. The fitted models share many observations, so their held-out losses are generally dependent. We cannot treat the k fold means as independent measurements and automatically divide their variance by k. Choose folds that represent the intended prediction setting. For example, the usual random-fold argument relies on observations being suitable for that exchange of roles.

A selection score has already helped make the choice

For each candidate we can compute a validation or cross-validation score. We then compare those scores and retain the smallest. This last operation changes how the selected score should be interpreted. It was not obtained for a candidate fixed independently of that comparison; it was one reason the candidate was chosen.

Return to the ridge table. The value 0.02588 selected $\lambda = 2$ on those two validation rows. Reporting it as the performance of the completed choice on untouched test data would give the same rows two different roles. A final test set is a separate set of observations reserved for evaluating the choice after the selection rule has finished.



The choice uses validation; the final score uses separate observations.

The training data produce candidate fits, and validation data help choose a setting. The chosen fitting rule may then be refitted on the development data. Final test responses enter only the last score; changing the choice after seeing that score would use them for selection.

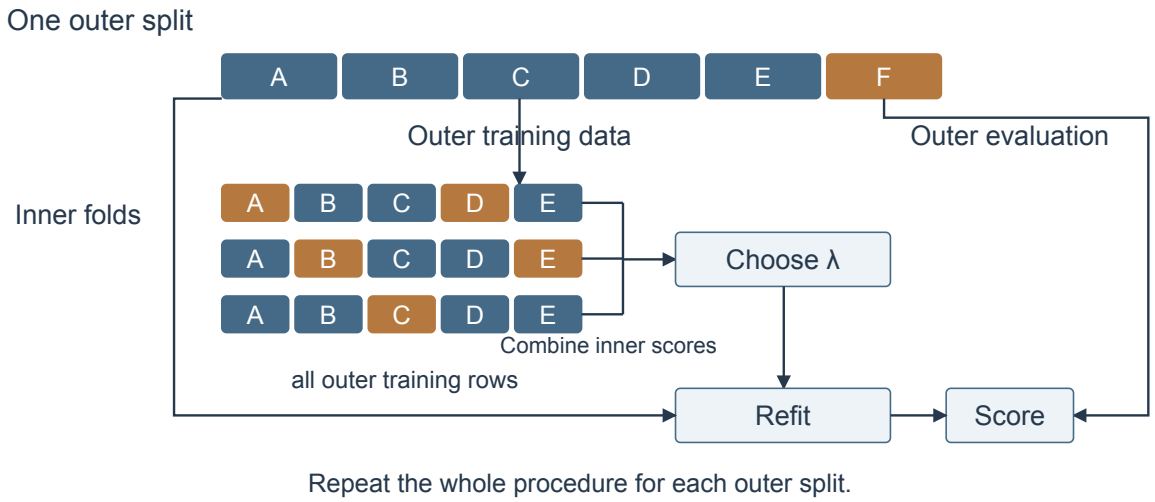
This distinction applies to the whole procedure. Choosing features, changing preprocessing, choosing λ, and choosing among model families can all use evidence. If they use a set of responses, those responses are no longer untouched final evaluation data. The issue is their actual use, not whether a file was named "test."

Sometimes we cannot reserve a large separate final test set. We can instead place the entire selection operation inside an outer cross-validation loop.

Definition: Nested cross-validation

For each outer split, set aside an outer evaluation fold. Use only the remaining outer training data to compare candidates by an inner validation or cross-validation rule. Refit the selected candidate on the outer training data, then score the untouched outer fold. Repeat for the other outer splits and combine their held-out losses.

The outer score evaluates the procedure "select, then fit" at the outer training-set size. The setting selected by the inner rule may differ between outer splits.



Fold F is set aside before the inner comparisons. All inner folds come from A–E. After choosing a setting, the rule is refitted using A–E and scored on F. Rotate the outer fold and repeat the entire operation.

The inner winner need not be the same in every outer split. That variation belongs to the selection procedure being evaluated. Forcing all outer splits to use a setting chosen with the complete dataset would allow the outer responses to influence the choice.

After the outer evaluation, the same selection rule can be applied to all development data to construct a final fit. That fit has more training observations than an outer fit. The outer result is evidence about the procedure; it is not an exact, universally unbiased risk estimate for that different final fit. Keeping the training size and prediction target visible avoids this confusion.

We now know where a candidate score and a final evaluation score belong. We can improve the calculation inside this protocol without changing its information boundary. In ridge regression, an especially useful case is to hold out just one observation at a time.

Calculate a prediction that did not use its own response

What does leaving one observation out change?

In ordinary cross-validation, a fold may contain several observations. At the smallest possible fold size, each fold contains one observation. A fit then uses almost all the available data, but its prediction for the omitted row still excludes that row's response.

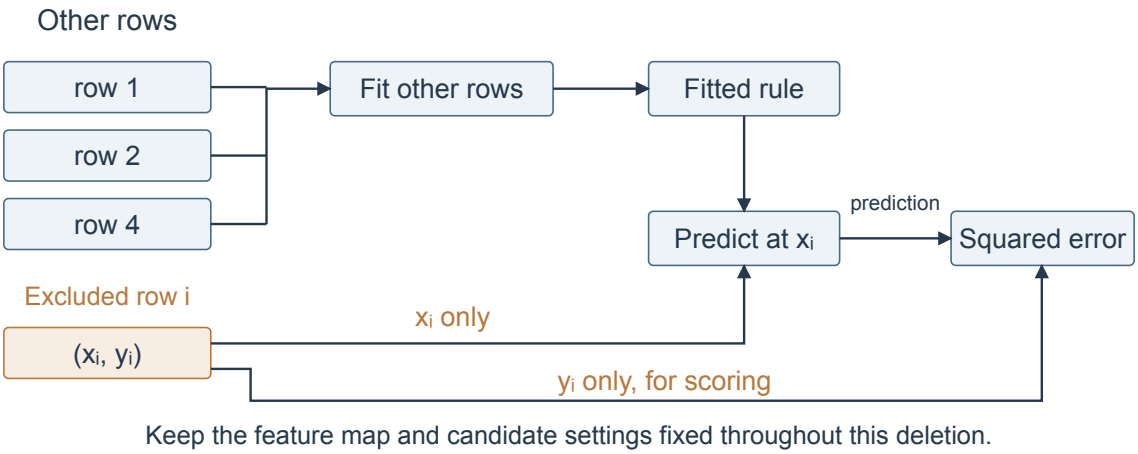
Definition: Leave-one-out cross-validation

For each observation i, remove the pair (x_i, y_i) and fit the declared rule to the other $n-1$ observations. Write the resulting predictor as $\hat{f}^{-i}$. **Leave-one-out cross-validation**, or **LOOCV**, uses the score

$$\mathrm{CV}_{\mathrm{LOO}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}^{-i}(x_i)\right)^2.$$

The model settings are fixed during this calculation. The fitted coefficients may change when the training row is removed.

The phrase "remove the pair" describes its role in fitting. We still need x_i to ask for a prediction, and we still need y_i to judge that prediction. Sending the feature vector to a fitted model does not train the model. Sending the response into coefficient fitting would do so.



Follow the omitted row in two directions. Its features enter prediction after the model is fitted. Its response enters only the final comparison. All other rows provide the fitting data.

This distinction also explains why setting y_i to zero is not deletion. A row with a zero response remains a training observation and still contributes a squared-loss term. Deletion removes the whole term. We will see that it changes both sides of the normal equations.

LOOCV supplies n predictions, each obtained from $n-1$ observations. Its training size is therefore close to that of a full-data fit, but not identical. The predictions also share many training observations. We should not treat their errors as independent measurements, or claim that this choice of folds always has the smallest variance. Here our immediate question is computational: must we really fit the model n times?

Name the quantity before simplifying it

Let $\hat{y}_i = \hat{f}(x_i)$ be the prediction from the full-data fit. Its ordinary residual is $e_i = y_i - \hat{y}_i$. The deleted prediction uses a different fit, so give its residual a different symbol:

$$d_i = y_i - \hat{f}^{-i}(x_i).$$

Definition: Prediction residual sum of squares

The **prediction residual sum of squares**, abbreviated **PRESS**, is the sum of squared leave-one-out residuals:

$$\text{PRESS} = \sum_{i=1}^{n} d_i^2.$$

For squared loss, $\text{CV}_{\text{LOO}} = \frac{\text{PRESS}}{n}$. PRESS is a total; the LOOCV score is a mean. Both use deleted predictions, not the full-data fitted values.

The definition already tells us how to compute PRESS: refit after each deletion, predict the omitted response, square the residual, and add. A shortcut must produce those same residuals for the same fitting rule. It cannot change the experiment merely to make the calculation faster.

Return to the four training rows in our example. We will delete one of these four at a time with $\lambda = 2$. The two validation rows from the earlier comparison stay outside this experiment. Holding λ fixed here lets us study one candidate. If a larger procedure chooses λ using data, evaluating that procedure still requires the separation established in the first section.

See how the full fit uses the responses

For a general design matrix X with n rows and p columns, row i is x_i^T. The response vector y has length n, and the coefficient vector β has length p. Use the summed-loss ridge objective

$$\|y - X\beta\|^2 + \lambda\|\beta\|^2.$$

Every coefficient is penalized in this formula. We keep the feature construction and $\lambda > 0$ fixed when deleting a row. In particular, the formula does not silently re-estimate a feature transformation from all responses.

The normal equations give

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y, \quad \hat{y} = X\hat{\beta} = H_\lambda y,$$

where

$$H_\lambda = X(X^T X + \lambda I)^{-1} X^T.$$

The identity matrix I is p by p; H_λ is n by n. Once X and λ are fixed, this matrix converts the observed responses into the fitted responses.

Definition: Linear smoother

A fitting rule is a **linear smoother in the responses** if its fitted vector can be written as $\hat{y} = Hy$, where H does not depend on y when the design and model settings are fixed. The matrix H is called the **smoother matrix** or **hat matrix**.

"Linear" here refers to the dependence on y. It does not require the plotted prediction to be a straight line in the original feature. For example, fixed polynomial features may be included in X. Their construction must remain fixed for the matrix expression being used.

The ith row of H gives

$$\hat{y}_i = \sum_{j=1}^{n} h_{ij} y_j.$$

Thus h_{ii} is the coefficient of observation i's own response in its full-data fitted value. That coefficient is precisely the kind of self-use we need to account for when moving to a deleted prediction.

Definition: Leverage and effective degrees of freedom

For this smoother, the diagonal entry h_{ii} is the **leverage** of row i. With the design and settings fixed, changing only y_i by Δ changes $\hat{y}_i$ by $h_{ii}\Delta$.

The trace-based **effective degrees of freedom** is

$$\mathrm{df} = \mathrm{tr}(H) = \sum_{i=1}^{n} h_{ii}.$$

It measures the total diagonal response sensitivity of the fitted vector.

For full-column-rank ordinary least squares, this trace equals p. Ridge shrinks the fit, and its trace can be smaller than p. This is useful language for the flexibility of a smoother. Later, an information criterion will count estimated likelihood parameters. That count has a different definition; we will not substitute a trace into its formula without a separate justification.

Derive the deletion shortcut from the fitting rule

We now have the objects needed for the calculation. Write

$$A = X^T X + \lambda I, \quad b = X^T y.$$

Then the full-data equations are $A\hat{\beta} = b$. Row i contributes $x_i x_i^T$ to $X^T X$ and $x_i y_i$ to $X^T y$. Removing the row therefore gives

$$(A - x_i x_i^T)\hat{\beta}^{-i} = b - x_i y_i.$$

This equation uses the same numerical λ in the summed-loss objective. Only the indicated observation term has disappeared.

Move the deleted row's matrix contribution to the right:

$$A\hat{\beta}^{-i} = b - x_i y_i + x_i x_i^T \hat{\beta}^{-i}.$$

Subtract this equation from $A\hat{\beta} = b$. The expression in parentheses is exactly the deleted residual:

$$A\left(\hat{\beta} - \hat{\beta}^{-i}\right) = x_i\left(y_i - x_i^T \hat{\beta}^{-i}\right) = x_i d_i.$$

Multiplying by A^{-1} gives the coefficient change. Multiplying next by x_i^T gives the prediction change at the omitted feature vector:

$$\hat{\beta} - \hat{\beta}^{-i} = A^{-1} x_i d_i,$$

$$x_i^T \hat{\beta} - x_i^T \hat{\beta}^{-i} = x_i^T A^{-1} x_i d_i = h_{ii} d_i.$$

The full-data residual can now be related to the deleted residual. Insert the deleted prediction between the response and the full prediction:

$$e_i = y_i - x_i^T \hat{\beta} = \left(y_i - x_i^T \hat{\beta}^{-i}\right) - \left(x_i^T \hat{\beta} - x_i^T \hat{\beta}^{-i}\right) = d_i - h_{ii} d_i = (1 - h_{ii}) d_i.$$

The subtraction has a simple meaning. The full fit has used observation i and has changed its prediction by $h_{ii} d_i$ relative to the deleted fit. Correcting that self-use recovers the residual from a fit that excluded the observation.

Result: Fixed-ridge and OLS deletion identity

Use fixed features and a fixed summed-loss penalty $\lambda > 0$, with every coefficient penalized as above. For the corresponding full and deleted ridge fits,

$$d_i = \frac{e_i}{1 - h_{ii}}, \quad \text{PRESS} = \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}} \right)^2.$$

The ridge full and deleted matrices are positive definite, so both fits exist and $1 - h_{ii} > 0$.

For ordinary least squares, $\lambda = 0$. The same identity holds when the full design and every deleted design have full column rank. These conditions ensure the required inverses and nonzero denominators.

To see why positive ridge penalty protects the denominator, write $B = A - x_i x_i^T$ and $t = x_i^T B^{-1} x_i \geq 0$. Solving $Av = x_i$ gives $v = B^{-1} \frac{x_i}{1+t}$, so $h_{ii} = \frac{t}{1+t} < 1$. Thus the division is well-defined for this ridge rule.

The result is more specific than the statement "the fit is a linear smoother." Response-linearity describes the full-data map $y \mapsto Hy$. It does not specify how the rule changes after a row is removed. Our identity was proved from compatible full and deleted normal equations. A different smoother needs its own argument before this shortcut can be used.

Check the formula against actual refitting

For the four-row ridge example, $\sum x_i^2 = 30$, $\sum x_i y_i = 27$, and $\lambda = 2$. Hence $A = 32$, $\hat{\beta} = \frac{27}{32} = 0.84375$, and $h_{ii} = \frac{x_i^2}{32}$. The following values have already been calculated; the task is to compare their meanings.

Worked example: One full fit reproduces four deleted fits

The direct deleted coefficient is

$$\hat{\beta}^{-i} = \frac{27 - x_i y_i}{32 - x_i^2}.$$

The table compares its prediction residual with the shortcut. Displayed values are rounded.

Row i	Full residual e_i	Leverage h_{ii}	Direct deleted residual	Shortcut $\frac{e_i}{1-h_{ii}}$
1	0.15625	0.03125	0.16129	0.16129
2	0.31250	0.12500	0.35714	0.35714
3	−0.53125	0.28125	−0.73913	−0.73913
4	0.62500	0.50000	1.25000	1.25000

Look at row 4. Deleting $(4, 4)$ gives $\hat{\beta}^{-4} = \frac{11}{16} = 0.6875$. Its prediction is $4(0.6875) = 2.75$, so the deleted residual is $4 - 2.75 = 1.25$. The full fit predicts 3.375, giving residual 0.625. The shortcut divides that residual by $1 - 0.5$, also giving 1.25.

Across all four rows,

$$\text{PRESS} \approx 2.26238, \quad \text{CV}_{\text{LOO}} \approx 0.56559.$$

> The full-data training mean squared loss was only 0.19873. That smaller value used each response in its own fit. The deleted score asks the prediction question under the declared four-row experiment.

Row 4 has the greatest leverage here, and deleting it changes the prediction substantially. Leverage alone does not determine the deleted residual: the numerator e_i also matters. If a row has zero full residual and a valid nonzero denominator, its deleted residual is zero as well. A large leverage is therefore a reason to inspect the residual correction, not a declaration that the observation must have a large prediction error.

The trace in this example is $\frac{30}{32} = 0.9375$, below the OLS value of one for this one-coefficient model. This agrees with the reduced response sensitivity caused by ridge shrinkage. It does not mean that the ridge model literally estimates a fractional number of coefficients.

Keep the computation and the experiment aligned

The shortcut saves repeated coefficient fitting. It does not choose a penalty, repair data leakage, or evaluate a selection rule by itself. If we compute it for several fixed penalties and choose the lowest score, that score has participated in selection. The earlier outer-evaluation argument still applies.

The penalty scale also matters. The average-loss objective

$$\frac{1}{n}\|y - X\beta\|^2 + \lambda_{\text{mean}}\|\beta\|^2$$

has the same minimizer as a summed-loss objective with $\lambda_{\text{sum}} = n\lambda_{\text{mean}}$. After deletion there are $n-1$ terms. Keeping the numerical average-loss penalty fixed then produces summed penalty $(n-1)\lambda_{\text{mean}}$, not the original $n\lambda_{\text{mean}}$. Our derivation held the summed penalty fixed. A software call that changes this scale is fitting a different deleted rule.

Finally, suppose a feature transformation is estimated from training data. An honest deleted fit must estimate it without row i too. If that changes the design representation, the simple fixed-X argument above does not automatically compute the complete procedure's deleted predictions. Direct refitting remains the definition; a shortcut is acceptable only when it reproduces that definition.

The data protocol decides which observations may influence each operation. The algebra makes the specified fits cheaper to calculate. Could we obtain useful selection evidence from a training fit itself? For probability models, a correction for the favorable training fit offers another approach.

Use likelihood to compare predictive models

Change the model description, and say what changes

Cross-validation evaluates a declared learning rule by holding observations out. It can use squared loss for ridge regression, but it does not require that particular model or loss. We now consider candidates that describe a conditional probability distribution for the response. Such a model tells us more than a predicted mean: it assigns probabilities or densities to possible responses at a given feature vector.

This section uses ordinary maximum-likelihood fits in regular, fixed-dimensional models. We are changing from the penalized ridge calculation to a different setting. The likelihood parameter count introduced below is not automatically the ridge trace from the previous section.

Definition: Conditional probability model and parameter count

A **conditional probability model** specifies $p_\theta(y \mid x)$ for each parameter vector θ in a declared parameter space. It describes the distribution of the response at given features.

The **parameter count** q is the number of freely estimated scalar parameters in that candidate model. Parameters fixed in advance are not counted. Estimated distribution parameters, such as a noise variance, are counted along with parameters of the mean.

For example, a Gaussian response model with a constant mean and unknown variance estimates two parameters: the mean and the variance. A Gaussian model with mean $a + bx$ and unknown variance estimates three: a, b, and the variance. Even if we mainly care about predicting the mean, the likelihood uses the fitted variance too.

Assume the observations are independent under the sampling model. Their conditional probabilities or densities multiply. A model that assigns greater density to the observed responses has a larger likelihood, with the feature values held fixed in that comparison.

Definition: Likelihood, log-likelihood, and maximum likelihood

For data $S = ((x_i, y_i))_{i=1}^n$, the **conditional likelihood** is

$$L_S(\theta) = \prod_{i=1}^{n} p_\theta(y_i \mid x_i).$$

Its logarithm, the **log-likelihood**, is

$$\ell_S(\theta) = \sum_{i=1}^{n} \log(p_\theta(y_i \mid x_i)).$$

A **maximum-likelihood estimate** (MLE), denoted $\hat{\theta}_S$, maximizes this log-likelihood over the candidate's parameter space. Write $\hat{\ell} = \ell_S\left(\hat{\theta}_S\right)$ for its fitted value.

The logarithm preserves the ranking because it is increasing. It also turns the product into a sum, so each observation has an additive contribution. This makes the connection to a prediction loss straightforward.

Definition: Log loss

The **log loss** of the probability model at one observed pair is

$$-\log(p_\theta(y \mid x)).$$

For a dataset of size n, its total log loss is $-\ell_S(\theta)$ and its mean log loss is $-\frac{\ell_S(\theta)}{n}$. Smaller loss means that the model assigns greater probability or density to the responses that actually occur.

A continuous density can exceed one, so a log loss need not be positive. That does not invalidate comparisons on the same response scale. What matters is that all candidates describe the same observed responses under a common probability or density convention.

Training likelihood and new-data likelihood have different roles

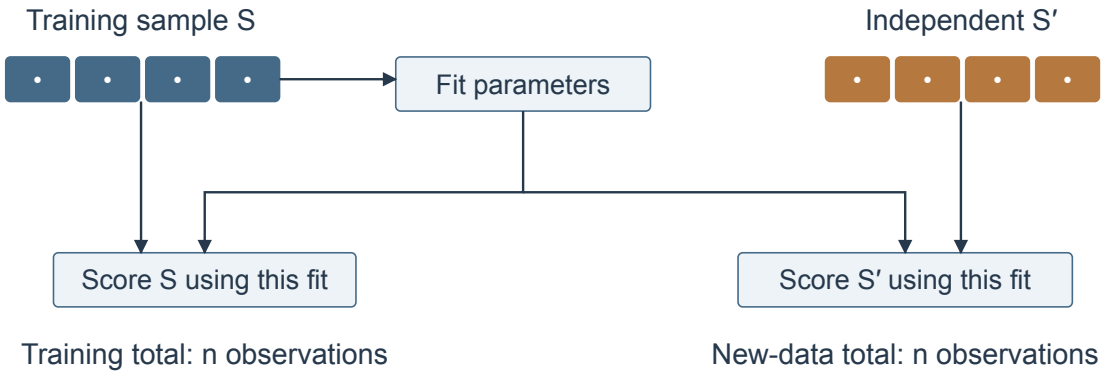
Maximum likelihood deliberately finds parameters that make the observed training responses look plausible. A model with more free parameters may improve that training fit. The useful question is whether the improvement survives on new responses.

Imagine an independent sample S' of the same size n from the same population. Fit the model once using S, then keep $\hat{\theta}_S$ fixed while scoring S'. Do not refit it on S'. The new sample plays the role of evaluation data, just as it did in the earlier data protocol.

The standard information-criterion scale is **twice total log loss**. On this scale the observed training score is $-2\ell_S\left(\hat{\theta}_S\right)$. The predictive target we want to understand is

$$E_S E_{S'}\left[-2\ell_{S'}\left(\hat{\theta}_S\right)\right].$$

The outer expectation averages over possible training samples and their fitted parameters. The inner expectation averages over independent new samples, with that fitted model held fixed. The target concerns repeated use of the fitting rule, not just one unusually favorable dataset.



The same fitted parameters are used in both scores. Only the observations being scored change. Each score totals n terms on the -2 log-likelihood scale; taking a mean would divide the entire score by n.

Using the same number of observations makes the total scores comparable. If we instead want mean log loss, we divide a criterion on the twice-total scale by $2n$, including its correction term. Dividing only the fitted-loss term changes the balance between fit and complexity.

Why should a training score require a correction? The fitting operation has searched for parameters favorable to S. The independent sample S' has not helped with that search. A model can therefore fit its training responses better than it predicts independent responses. The **optimism** of the training score means this favorable difference, measured in expectation on a specified loss scale.

Connect predictive log loss to the unknown distribution

A good probability model should resemble the true conditional distribution, denoted $p_*(y \mid x)$. We do not know this distribution, so we cannot directly compare every candidate to it. A mathematical discrepancy nevertheless clarifies what expected log loss rewards.

Definition: Conditional Kullback–Leibler divergence

At a fixed feature value x, the **Kullback–Leibler divergence** from the true response distribution to a model is

$$\mathrm{KL}_x(p_* \parallel p_\theta) = \int p_*(y \mid x) \log \frac{p_*(y \mid x)}{p_\theta(y \mid x)} \, \mathrm{d}y.$$

For discrete responses, replace the integral by a sum. Averaging this quantity over the feature distribution gives the conditional divergence relevant to future observations:

$$E_X[\mathrm{KL}_X(p_* \parallel p_\theta)].$$

These expressions require suitable support and integrability; a model assigning zero probability where the truth has positive probability can have infinite divergence.

Split the logarithm of a ratio into a difference. The average divergence becomes

$$E_X[\mathrm{KL}_X(p_* \parallel p_\theta)] = E_{X,Y}[\log(p_*(Y \mid X))] + E_{X,Y}[-\log(p_\theta(Y \mid X))].$$

The first term depends only on the true distribution. The second is the model's expected log loss. Thus, when these quantities are finite, minimizing expected log loss also minimizes this divergence. We do not need to know the truth-only term to compare candidates.

This identity explains the predictive interpretation of likelihood criteria. The aim is not simply to make training likelihood large. It is to use a training-data calculation to compare fitted models by their expected new-data log loss, and therefore by their average discrepancy from the truth. For a fitted parameter $\hat{\theta}_S$, we also average over the training sample that produced it.

The identity does not yet tell us how much to correct the training score. That amount depends on the fitting setting and assumptions. The familiar correction below comes from ordinary regular maximum-likelihood estimation; it is not a consequence of the KL definition alone.

Add a correction on the same scale

Definition: Akaike information criterion

For an ordinary maximum-likelihood fit with q freely estimated parameters, including any estimated variance, the **Akaike information criterion** is

$$\mathrm{AIC} = -2\hat{\ell} + 2q.$$

Here $\hat{\ell}$ is the total training log-likelihood evaluated at the maximum-likelihood estimate, and n is the number of observations. Use natural logarithms.

Compare candidates fitted to the same observations using comparable likelihoods. Smaller AIC is preferred. On the mean-log-loss scale, the same ranking uses

$$\frac{\mathrm{AIC}}{2n} = -\frac{\hat{\ell}}{n} + \frac{q}{n}.$$

The first term rewards a good training fit. The second charges for the optimism introduced by estimating parameters. Both terms belong to the same total-score scale. The number $2q$ is not a generic penalty that can be attached to any error measure.

The result below uses several conditions. A model is **identifiable** when distinct parameter values specify distinct distributions. A true parameter is **interior** when it is not on the boundary of the allowed parameter space. A **regular** estimation setting excludes singular or otherwise unstable parameter estimation; smoothness alone is not enough.

Result: The predictive interpretation of the AIC correction

Consider a correctly specified, regular, identifiable probability model fitted by ordinary maximum likelihood. Keep its parameter dimension q fixed as n grows. Assume the true parameter is an interior point, the likelihood is sufficiently smooth and well behaved, and the required expectations are finite with suitable large-sample expectation control.

The remainder $o(1)$ tends to zero as the sample size grows under the stated conditions.

Under these conditions, the leading expected difference between independent same-size new-data score and fitted training score, on the -2 log-likelihood scale, is $2q$:

$$E_S E_{S'}\left[-2\ell_{S'}\left(\hat{\theta}_S\right)\right] = E_S\left[-2\ell_S\left(\hat{\theta}_S\right)\right] + 2q + o(1).$$

Thus AIC supplies a large-sample estimate of this expected predictive score. This is an expectation statement, not an exact equality for each dataset.

The expectation conditions matter because convergence of fitted parameters alone does not automatically justify averaging their losses.

The formula helps interpret AIC without asking us to derive a new asymptotic theory in this lesson. It says that training fit is favorable by about $2q$ on average in this setting. It does not say the realized prediction loss equals AIC, or that AIC is an upper confidence bound. Selecting the smallest observed AIC also does not remove the uncertainty of comparing candidates.

The ordinary $2q$ correction is not guaranteed by this result for arbitrary misspecified models, penalized fits, boundary parameters, or singular models. In particular, the ridge smoother from the previous section does not become an ordinary q-parameter MLE merely because its fitted values are linear in y. These are conditions on the interpretation at hand, not alternative definitions of the score.

Read a completed comparison

The following is a separate illustrative setting with $n = 100$. It is not a continuation of the four-row ridge data. Both candidates are Gaussian conditional models fitted to the same responses by ordinary maximum likelihood. Their variance is estimated in both cases.

Worked example: A better training likelihood must earn its extra parameter

The fitted likelihood summaries are supplied:

Candidate mean	Parameter count q	$-2\hat{\ell}$	AIC
Constant mean a	2	100	104
Linear mean $a + bx$	3	97	103

The constant-mean model estimates a and a variance. The linear-mean model estimates a, b, and a variance. Adding the slope improves the fitted score by $100 - 97 = 3$. Its extra AIC

penalty is $2(3-2)=2$. The fit improvement exceeds the extra penalty, so AIC prefers the linear-mean candidate by one unit.

This is a decision from the supplied evidence. A difference of one is not a proof that the linear model has lower population prediction loss. The table also does not say that the mean squared error improved by three; the fitted score is on the -2 log-likelihood scale.

The scale distinction can be made explicit for a Gaussian model with known common variance σ^2. If its mean predictions have residual sum of squares RSS, then

$$-2\ell = n\log(2\pi\sigma^2) + \frac{\mathrm{RSS}}{\sigma^2}.$$

Even here, the fitted term is a scaled residual sum plus a constant, not raw mean squared error. If the variance is estimated, its fitted value also enters the likelihood and its parameter must be counted. We should start from the stated likelihood instead of attaching $2q$ to an unrelated score.

Cross-validation obtains predictive evidence by leaving observations out. AIC uses the fitted likelihood and an assumption-based correction for expected optimism. In both cases, we must identify the score and the fitting operation. But prediction is not the only possible selection goal. The same fitted likelihood can also help us compare models as descriptions of the population distribution.

Distinguish predictive fit from model identification

Use the same fit evidence with a different penalty

Suppose our scientific question is whether a slope is needed in the conditional model. We may want to identify a model that represents the true distribution, rather than choose whichever fitted candidate has the smallest estimated predictive loss at the present sample size. These aims are related, but they are not identical.

Definition: Bayesian information criterion

For a regular maximum-likelihood candidate with q freely estimated parameters and n observations, the **Bayesian information criterion** is

$$\mathrm{BIC} = -2\hat{\ell} + q\log n.$$

Here $\hat{\ell}$ is again the total training log-likelihood evaluated at the maximum-likelihood estimate. The count q includes any estimated variance. Use the natural logarithm. Compare candidates on the same observations with comparable likelihoods; smaller BIC is preferred.

The fitted term is the same as in AIC. The charge per parameter changes from 2 to $\log n$. When $n > e^2$, BIC charges more per additional parameter than AIC. It can therefore select a simpler model even when AIC prefers the more flexible one. This is a consequence of the formulas and their selection goals, not an arithmetic inconsistency.

BIC has a large-sample connection to model evidence under suitable assumptions. Here we use its stated criterion and a qualified model-identification property. Computing the criterion requires the fitted log-likelihood, parameter count, and sample size; it does not require a new integration calculation in this tutorial.

Worked example: The same likelihood table can support different choices

Keep the two Gaussian candidates and $n = 100$ from the preceding example. Since $\log 100 \approx 4.60517$, the completed comparison is

Candidate	$-2\hat{\ell}$	AIC	BIC
Constant mean, $q = 2$	100	104	109.21034
Linear mean, $q = 3$	97	103	110.81551

Adding the slope improves fitted score by three, but costs about 4.605 under BIC. BIC therefore prefers the constant-mean model. AIC charges only two for the extra parameter and prefers the linear-mean model. Each comparison stays within one column. An AIC value of 103 and a BIC value of 109.21 are not competing estimates on which to choose between criteria.

State what consistency does and does not promise

To explain the model-identification property, first make the candidate structure precise. A collection is **nested** when each smaller model is contained in the next larger model. For example, the constant-mean Gaussian model lies inside the linear-mean Gaussian model by setting $b = 0$. The larger model can then represent every distribution represented by the smaller one.

Definition: Smallest true candidate and selection consistency

A candidate is **true** if it contains the population distribution. In a nested collection, the **smallest true candidate** is the least complex candidate that contains that distribution.

A model-selection rule is **consistent** for that candidate if the probability that it selects the candidate tends to one as the sample size tends to infinity.

Result: A qualified BIC model-identification guarantee

Consider a fixed finite nested collection of regular, identifiable, fixed-dimensional candidate models, with a unique smallest true candidate. Under the usual likelihood separation, estimation, and large-sample regularity conditions for these candidates, BIC selects that smallest true candidate with probability tending to one as n grows.

The two parts of the setting matter. A smaller false model cannot represent the population distribution; enough data can reveal its loss of fit. A larger true model contains extra parameters that are unnecessary for representing the truth. The growing parameter penalty helps distinguish that larger model from the smallest true one. This is an interpretation of the guarantee, not a replacement for its assumptions.

Consistency is a statement about a sequence of growing samples. It does not promise that BIC chooses correctly for our particular sample of 100 observations. If none of the candidates contains the true distribution, there is no smallest true candidate for this statement to identify. Nor does model-selection consistency imply that BIC always gives the best finite-sample predictions.

AIC's predictive target and BIC's qualified identification property now explain their disagreement in the table. Before asking which criterion to use, we must state the selection goal and

check whether the setting supports that interpretation. Neither name removes the need to evaluate a final predictive procedure appropriately.

Make the whole decision explicit

Match the evidence to the question

We began with a ridge parameter and a prediction goal. The important decision was not simply which formula to calculate. We needed to distinguish coefficient fitting, candidate selection, and evaluation of the completed procedure. The likelihood criteria added another distinction: predictive score and identification of a smallest true model are different targets.

The table below is a map for choosing evidence. Its rows describe different operations, so the numerical values from different rows should not be ranked against one another.

Evidence	Question it helps answer	What must be kept visible
Holdout or k-fold score	How does a declared candidate predict observations excluded from its fit?	Training size, loss, fold roles, and whether the score also chooses a candidate.
Nested cross-validation	How does the complete selection-and-fitting rule predict excluded outer observations?	All selection remains inside each outer training set.
PRESS shortcut	Can we compute specified leave-one-out predictions without repeated fits?	Fixed design and summed ridge penalty; compatible deleted fit and valid denominators.
AIC	Which candidate has the lower estimated log loss on new data?	Common likelihood scale, fitted parameter count, and regular MLE assumptions.
BIC	Which candidate is favored by a stronger large-sample complexity penalty?	Common likelihood scale; the smallest-true-model guarantee needs its stated setting.

For a new task, begin by naming the object that will be used on future observations. Is it one fixed candidate, or a rule that searches among candidates? Then name the intended loss or identification goal. Only after those decisions can we tell whether a score has the right interpretation.

A complete prediction decision

> **Worked example: Choose a ridge penalty and evaluate the procedure**
>
> Suppose the aim is to predict a real-valued response, with mean squared loss as the criterion. The available development data will be used to compare a declared set of ridge penalties. A separate final test set is available.
>
> First specify the candidate penalties, the loss, the fold construction, and any feature preparation. Within the development data, compare candidates by cross-validation. Any learned feature preparation is fitted using only the training portion of each split. Select the penalty using those scores.

> Next refit the selected rule using the development data. Evaluate that completed fit once on the independent final test set. The test responses have not chosen the penalty, feature preparation, or comparison rule. Report the test mean squared loss as evidence about this completed fit's predictions, together with the relevant sample sizes and uncertainty.
>
> If a separate test set is unavailable and the task is to evaluate the selection rule, use an outer cross-validation loop. Repeat the whole inner selection inside each outer training set. The resulting outer score concerns the procedure trained at the outer training size. It does not become a test score for a different full-data fit merely by renaming it.

Where can PRESS help in this workflow? It can accelerate the inner comparison when each candidate really is a fixed-design, fixed-summed-penalty ridge rule satisfying the identity. It replaces repeated fitting in that calculation. It does not replace the outer information boundary.

Our four-row example illustrates the same distinction at a small scale. Its validation table selected $\lambda = 2$ from three candidates. Its later PRESS calculation examined deletion for that fixed candidate on the original four rows. The calculated 0.56559 is not a new independent evaluation of the earlier choice. The datasets and operations must be traced, even when every number is correct.

A complete likelihood-model decision

> **Worked example: Choose a criterion by the intended use**
>
> Suppose two ordinary Gaussian regression models have been fitted to the same 100 observations. Both estimate their noise variance, so the constant-mean and linear-mean parameter counts are two and three. The supplied likelihood scores are 100 and 97 on the $-2\hat{\ell}$ scale.
>
> If the stated aim is predictive log loss and the regular MLE interpretation is appropriate, compare AIC: 104 against 103. This evidence favors the linear-mean candidate. Describe that result as a criterion-based preference, not a proof about its unknown population loss. If final predictive performance must be reported after selection, preserve an appropriate independent evaluation boundary.
>
> If the stated aim is identifying the smallest true candidate, first examine whether the fixed nested candidate collection and regularity assumptions are credible. Under that setting, BIC is aligned with the identification goal. Its values 109.21 and 110.82 favor the constant-mean candidate for the supplied sample. Its consistency property concerns increasing sample size; it does not certify this single decision.

The fitted likelihood evidence is the same in these two decisions. The target changes, and so does the penalty. State the purpose before interpreting the scores; choosing the purpose afterward would let a preferred answer determine the comparison.

Transfer the data-role rule to preprocessing

Consider this proposed workflow: estimate feature means and standard deviations on all available data, standardize every row, and then run cross-validation on the transformed rows. Does the held-out fold remain completely outside the fitting procedure?

No. Its feature values have already influenced the estimated transformation. Even though its responses were not used, the transformation has crossed the intended training boundary. For ordinary evaluation of a learning procedure on new observations, fit the transformation within each training split. Apply those fitted transformation parameters to the held-out features,

then predict and score. In nested cross-validation, the same rule applies at both levels: inner validation data do not fit inner transformations, and outer evaluation data do not fit any transformation used in the outer training procedure.

This answer follows from the earlier row-flow diagram. A new observation may supply its features to a transformation and predictor that have already been fitted. It should not help estimate that transformation when the intended evaluation is of a training-only procedure. Procedures explicitly designed to use an unlabeled evaluation sample would be a different declared learning setting.

That repair may also change which algebraic shortcut is available. If every deletion re-estimates a transformation, the design matrix changes beyond simply removing one fixed row. We must then refit the complete rule or establish a compatible shortcut. Computational convenience cannot decide which data are allowed to influence the model.

Connect to the earlier complexity principle

Earlier structural risk minimization compared empirical fit with a justified complexity allowance over declared model classes. Its risk-control statement required the assumptions and confidence allocation of that bound. The present methods share the motivation that training fit alone is insufficient, but their formulas and guarantees serve different purposes.

A cross-validation score measures held-out loss. AIC applies an expected-optimism correction on a likelihood scale. BIC uses a sample-size-dependent penalty with a qualified model-identification interpretation. An SRM upper bound controls a risk under its stated conditions. Calling all four "fit plus penalty" would conceal the evidence that makes each one meaningful. We cannot exchange their penalties or attach a confidence guarantee to a criterion merely because its formula contains a complexity term.

The useful common habit is to make the comparison rule explicit before using the data: state candidates, fitting operations, score scale, selection goal, and evaluation protocol. If a procedure searches over these choices as well, those searches belong inside the selection stage being evaluated.

Read the whole argument in standard terms

The central distinction is between **selection** and **evaluation**. Holdout and k-fold cross-validation provide excluded-observation predictions. Once their scores select a candidate, those scores have participated in the choice. A final test set or an outer cross-validation loop supplies a separate evaluation role.

LOOCV makes each held-out set a single row. PRESS totals the resulting squared residuals. The fixed-ridge or valid OLS identity computes those residuals from the full fit while preserving the deleted-fit experiment. Leverage explains the self-response correction, and the trace summarizes response sensitivity.

Likelihood models lead to another predictive loss: negative log probability or density. AIC corrects the fitted twice-total log loss by $2q$ under its regular MLE interpretation. BIC changes the parameter charge to $q \log n$ and supports a qualified smallest-true-model selection goal. The same fitted data can therefore yield different choices for different purposes.

To interpret a model-selection result, finish this sentence: "These observations were allowed to choose …; these other observations were used to evaluate …; the reported score measures …." If every blank names a precise operation and target, the numerical result can be interpreted. If a blank is unclear, trace the data roles before interpreting the number.